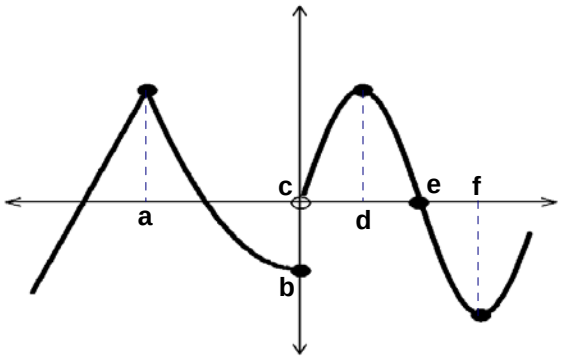
AP Calculus AB **Name**:_____

1) What are the three ways for a function to fail to be differentiable at a particular point $P(c, f(c))$:

a) _____

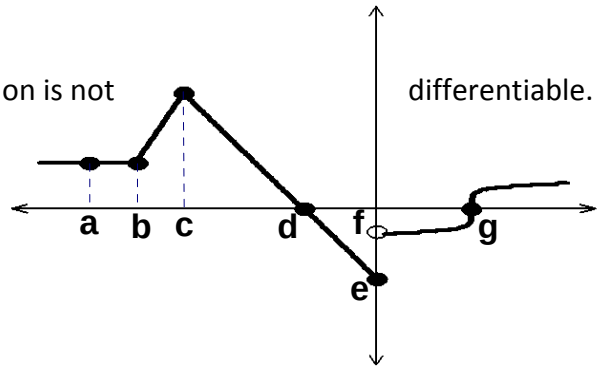
b) _____

c) _____



2) Use the graph to determine all x-values at which the function is not differentiable. Explain each answer.

3) Use the graph to determine all x-values at which the function is not differentiable. Explain each answer.

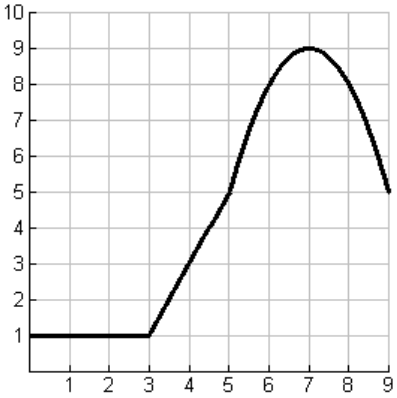


4) Let f(x) be the function whose graph is shown in the figure.

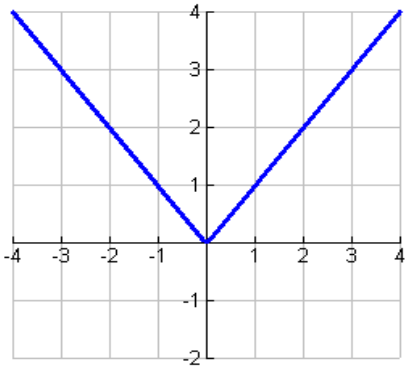
(a) Determine: $f'(1)$ =_____ , $f'(2)$ =_____, $f'(4)$ =_____, $f'(7)$ =_____.

(b) Estimate: $f'(6)$ =_____, $f'(8)$ =_____.

(c) Does $f'(3)$ exist?_____ Explain.



5) Use the given graph to estimate the value of each derivative. Then sketch the graph of f'.

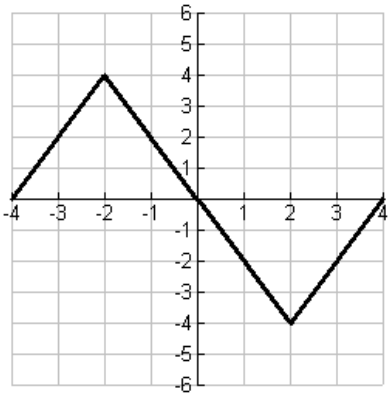


(a) $f'(-3) =$_____ (b) $f'(-2) =$_____

(c) $f'(-1) =$_____ (d) $f'(0) =$_____

(e) $f'(1) =$_____ (f) $f'(2) =$_____

(g) $f'(3) =$_____



6) Use the given graph to estimate the value of each derivative. Then sketch the graph of f'.

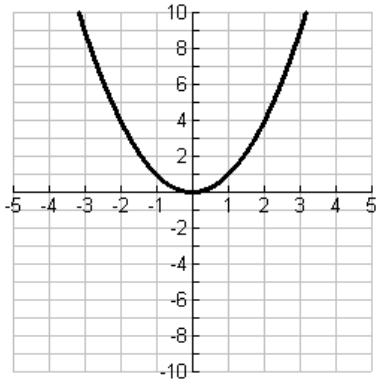
(a) $f'(-3) =$_____ (b) $f'(-2) =$_____

(c) $f'(1) =$_____ (d) $f'(0) =$_____

(e) $f'(1) =$_____ (f) $f'(2) =$ _____

(g) $f'(3) =$_____

7) Use the given graph to estimate the value of each derivative. Then sketch the graph of f'.

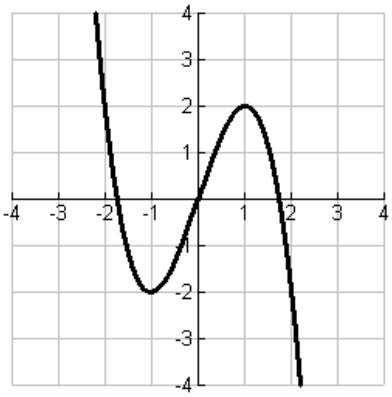


(a) $f'(-3) =$_____(b) $f'(-2) =$_____

(c) $f'(-1) =$_____(d) $f'(0) =$_____

(e) $f'(1) =$_____ (f) $f'(2) =$_____

(g) $f'(3) =$_____



8) Use the given graph to estimate the value of each derivative. Then sketch the graph of f'.

(a) $f'(-2) =$_____

(b) $f'(-1) =$_____

(c) $f'(0) =$_____

(d) $f'(1) =$_____

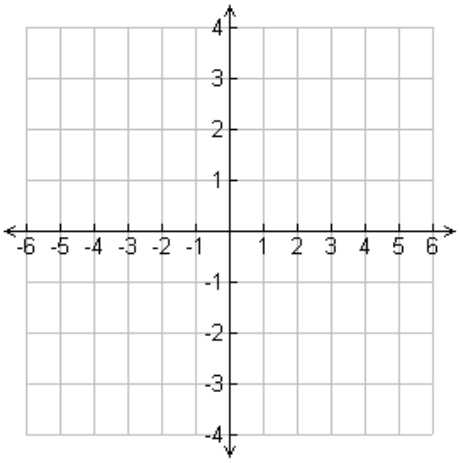
(e) $f'(2) =$_____

9) Let $\boldsymbol{f(x)=\sqrt[3]{x}}$.

(a) Sketch the graph of f(x).

(b) Find $\boldsymbol{f'(x)}$

(c) Show that ***f'*(0)** does not exist.



Definition of derivative

The derivative of $f(x)$ with respect to *x* is $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$ provided the limit exists. The process of calculating a derivative is called **differentiation**.

10) Find the derivative of the function using the definition of derivative: $\boldsymbol{f(x)=x^2-x}$

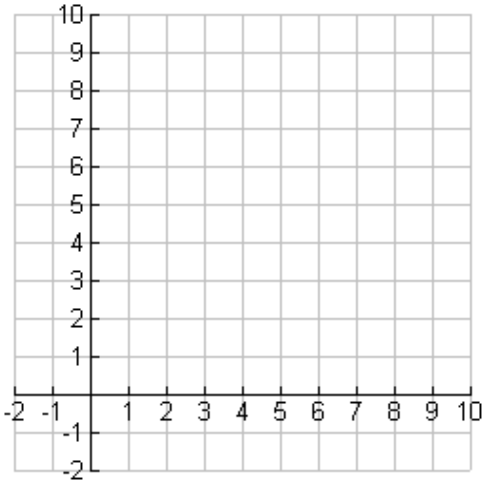
11) For $f(x)=\sqrt{x}$

(a) Find $f'(x)$ using the definition of derivative.

(b) State the domain of

i) $f(x)$.

ii) $f'(x)$.



(c) Sketch the graph of

i) $f(x)$
ii) $f'(x)$

12) Use the Power Rule to compute the derivative.

(a) $f(x) = x^3 + 4x^2 - 3x + 56$ (b) $f(x) = \frac{1}{x^3}$ (c) $f(x) = \sqrt[4]{x}$

13) Find $\frac{dy}{dx}$: $y = 6\cos(x) - 2\sin(x)$

14) Find the point on the curve $f(x) = x^2 + 3x - 7$ at which the slope of the tangent line is horizontal.

15) Find the point on the curve $f(x) = x^2 + 3x - 7$ at which the slope of the tangent line is equal to 4.

16) Given $f(x) = 4x - x^2$.

(a) Find the slope of the tangent line to the parabola $f(x) = 4x - x^2$ at the point (1, 3).

(b) Find the equation of the tangent line in part (a).

17) If $f(2)=6$ *and* $f'(x)=-3,$ find an equation of the tangent line when x = 2. Solve the equation for y.

18) If a ball is thrown into the air with a velocity of 40ft/sec, its height (in feet) after t seconds is given by $y=40t-16t^2$. Find the velocity when t = 2.